Continuum Mechanics
class #01
8/18/2026

We will follow the textbook

Gonzalez and Stuart, A First Course in Continuum Mechanics, Cambridge Univ. Press, 2008.

Chapter 01: Tensor Algebra

Read Section 1.1 on your own. We will start in Section 1.2.

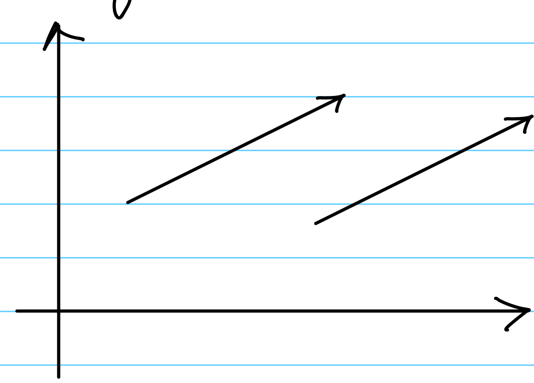
The book distinguishes between

$$\mathbb{E}^3 = \{\text{the set of points in Euclidean 3-space}\}$$

and

$$\mathcal{V} = \{\text{the space of vectors in Euclidean 3-space}\}$$

Points are immovable where as vectors (objects with a magnitude and direction) can be moved.



These two vectors are the same.

Section 1.2: Index Notation

Let us recall the canonical basis vectors in $\mathcal{V}$

$$\hat{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad \hat{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \qquad \hat{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

If we want to extract the j^{th} element from any vector

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

we write

$$[\vec{v}]_j = v_j, \qquad j = 1, 2, 3.$$

Thus,

$$[\hat{e}_i]_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}.$$

Try it out to convince yourself.

We have a special notation:

$$\delta_{i,j} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}, \qquad i, j \in \{1, 2, 3\},$$

is called the <u>Kronecker delta</u> symbol.

So

$$[\hat{e}_i]_j = \delta_{i,j}$$

Now, we can write

$$\vec{v} = \sum_{i=1}^{3} v_i \hat{e}_i$$

This can be expressed in short hand as

$$\vec{v} = v_i \hat{e}_i$$

using what is called the <u>summation convention</u>.

For any repeated index summation is implied. More about this in a moment.

More generally, let $\mathcal{F} = \{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ be a set of 3 mutually orthogonal vectors, ie,

$$\vec{e}_i \cdot \vec{e}_j = \delta_{i,j}, \qquad i,j = 1,2,3,$$

with the further property that

$$\vec{e}_1 \times \vec{e}_2 = \vec{e}_3,$$

$$\vec{e}_2 \times \vec{e}_3 = \vec{e}_1,$$

$$\vec{e}_3 \times \vec{e}_1 = \vec{e}_2.$$

$\mathcal{F}$ is called a <u>right-handed orthonormal basis</u>.

Given any vector $\vec{v} \in \mathcal{V}$, there are 3 unique scalars $v_1, v_2, v_3 \in \mathbb{R}$ such that

$$\vec{v} = v_1 \vec{e}_1 + v_2 \vec{e}_2 + v_3 \vec{e}_3$$

Of course these numbers change as the basis changes.

Let $\vec{a}, \vec{b} \in \mathcal{V}$ and suppose $\mathcal{F} = \{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ is a right-handed orthonormal basis. Write

$$\vec{a} = \sum_{i=1}^{3} a_i \vec{e}_i$$

$$\vec{b} = \sum_{i=1}^{3} b_i \vec{e}_i$$

$$\vec{a} \cdot \vec{b} = \left(\sum_{i=1}^{3} a_i \vec{e}_i \right) \cdot \left(\sum_{j=1}^{3} b_j \vec{e}_j \right)$$

$$= \sum_{i=1}^{3} \sum_{j=1}^{3} a_i b_j \vec{e}_i \cdot \vec{e}_j$$

$$= \sum_{i=1}^{3} \sum_{j=1}^{3} a_i b_j \delta_{ij}$$

$$= \sum_{i=1}^{3} a_i b_i$$

$$= a_i b_i,$$

with the summation convention.

We have already seen the Kronecker delta symbol:

$$\delta_{ij} = \vec{e}_i \cdot \vec{e}_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

where $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ is any frame (right-handed orthonormal basis).

We define the permutation symbol

$$\epsilon_{i,j,k} = (\vec{e}_i \times \vec{e}_j) \cdot \vec{e}_k$$

$$= \begin{cases} 1 & \text{if } ijk = 123, 231, 312 \\ -1 & \text{if } ijk = 321, 213, 132 \\ 0 & \text{otherwise (repeated indices)} \end{cases}$$

$\delta_{i,j}$ and $\epsilon_{i,j,k}$ are invariant to the choice of the orthonormal frame.

<u>Proposition</u>: For any indices $i, j, k \in \{1, 2, 3\}$

$$\epsilon_{i,j,k} = -\epsilon_{j,i,k},$$

$$\epsilon_{i,j,k} = -\epsilon_{i,k,j},$$

$$\epsilon_{i,j,k} = -\epsilon_{k,j,i}.$$

On the other hand

$$\epsilon_{i,j,k} = \epsilon_{j,k,i} = \epsilon_{k,i,j}$$

<u>Proposition</u>

$$\epsilon_{i,j,k} = \det\left(\begin{bmatrix} | & | & | \\ \vec{e}_i & \vec{e}_j & \vec{e}_k \\ | & | & | \end{bmatrix}\right)$$

Let's look at some further frame identities.

Proposition: Let $\{\vec{e}_i\}$ be any orthonormal frame. Then

$$\vec{e}_i = \delta_{ij}\vec{e}_j$$

and

$$\vec{e}_i \times \vec{e}_j = \epsilon_{i,j,k}\vec{e}_k$$

Proof: The summation convention is used in both of these identities. In particular

$$\vec{e}_i = \sum_{j=1}^{3} \delta_{i,j}\vec{e}_j$$

$$= \delta_{i,1}\vec{e}_1 + \delta_{i,2}\vec{e}_2 + \delta_{i,3}\vec{e}_3$$

$$= (\vec{e}_i\cdot\vec{e}_1)\vec{e}_1 + (\vec{e}_i\cdot\vec{e}_2)\vec{e}_3 + (\vec{e}_i\cdot\vec{e}_3)\vec{e}_3$$

We must take cases.

Case 1: $i=1$

$$\vec{e}_i\cdot\vec{e}_1 = 1, \quad \vec{e}_i\cdot\vec{e}_2 = 0, \quad \vec{e}_i\cdot\vec{e}_3 = 0.$$

$$\vec{e}_1 = \vec{e}_i \overset{?}{=} 1\vec{e}_1 + 0\vec{e}_2 + 0\vec{e}_3$$

$$= \vec{e}_1 \quad \checkmark$$

Case 2: $i=2$

$$\vec{e}_i\cdot\vec{e}_1 = 0, \quad \vec{e}_i\cdot\vec{e}_2 = 1, \quad \vec{e}_i\cdot\vec{e}_3 = 0$$

This yields

$$\vec{e}_2 = \vec{e}_i \overset{?}{=} 0\vec{e}_1 + 1\vec{e}_2 + 0\vec{e}_3$$
$$= \vec{e}_2 \checkmark$$

<u>Case 3</u>: This case is similar.

The second identity is left as an exercise. ///

Identity 1 is called the <u>transfer property</u> of $\delta_{i,j}$.

It holds in other forms as well:

$$v_i = \delta_{i,j} v_j, \qquad i \in \{1,2,3\},$$

for any vector $\vec{v}$.

<u>Exercise</u>: Show that, for any orthogonal frame $\mathcal{F} = \{\vec{e}_i\}$,

$$\vec{e}_i = \frac{1}{2} \epsilon_{i,j,k} \vec{e}_j \times \vec{e}_k$$

<u>Vector Operations in Component Form</u>

<u>Proposition</u>: Let $\mathcal{F} = \{\vec{e}_i\}$ be an orthonormal frame. Suppose that $\vec{a}, \vec{b}, \vec{c} \in \mathcal{V}$ are arbitrary and

$$\vec{a} = a_i \vec{e}_i, \quad \vec{b} = b_i \vec{e}_i, \qquad \vec{c} = c_i \vec{e}_i$$

Then

$$\vec{a} \cdot \vec{b} = a_i b_i$$

$$\vec{a} \times \vec{b} = a_i b_j \epsilon_{i,j,k} \vec{e}_k$$ (three implied sums)

(1.2) $$(\vec{a} \times \vec{b}) \cdot \vec{c} = \epsilon_{i,j,k} a_i b_j c_k$$ (three implied sums)

Proof: First

$$\vec{a} \cdot \vec{b} = (a_i \vec{e}_i) \cdot (b_j \vec{e}_j)$$

$$= a_i b_j \vec{e}_i \cdot \vec{e}_j$$

$$= a_i b_j \delta_{i,j}$$

$$= a_i b_i$$ (transfer property)

Next,

$$\vec{a} \times \vec{b} = (a_i \vec{e}_i) \times (b_j \vec{e}_j)$$ (two sums)

$$= a_i b_j \vec{e}_i \times \vec{e}_j$$ (two sums)

$$= a_i b_j \epsilon_{i,j,k} \vec{e}_k \quad //$$ (three sums)

Let's redo this with summation to see what we save by using the summation convention.

$$\vec{a} \times \vec{b} = \left(\sum_{i=1}^{3} a_i \vec{e}_i\right) \times \left(\sum_{j=1}^{3} b_j \vec{e}_j\right)$$

$$= \sum_{i=1}^{3} a_i \vec{e}_i \times \left(\sum_{j=1}^{3} b_j \vec{e}_j\right)$$

$$= \sum_{i=1}^{3} a_i \sum_{j=1}^{3} b_j \; \vec{e}_i \times \vec{e}_j$$

$$= \sum_{i=1}^{3} \sum_{j=1}^{3} a_i b_j \; \vec{e}_i \times \vec{e}_j$$

$$= \sum_{i=1}^{3} \sum_{j=1}^{3} a_i b_j \sum_{k=1}^{3} \epsilon_{i,j,k} \vec{e}_k$$

$$= \sum_{i=1}^{3} \sum_{j=1}^{3} a_i b_j \epsilon_{i,j,k} \vec{e}_k \quad //$$

For (1.2),

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = (a_i b_j \epsilon_{i,j,k} \vec{e}_k) \cdot (c_m e_m)$$

Here is the trick: set

$$\vec{d} = \vec{a} \times \vec{b}$$

Then

$$d_k = a_i b_j \epsilon_{i,j,k} , \qquad k = 1, 2, 3 .$$

(Do you see why this is true?) Thus, by the dot-product rule

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = d_k c_k \qquad (\text{one sum})$$

$$= a_i b_j \epsilon_{i,j,k} c_k$$

$$= a_i b_j c_k \epsilon_{i,j,k} \quad /// \quad (3 \text{ sums})$$

Note that

$$a_k b_k = a_r b_r = a_q b_q$$

These repeated indices (over which summation takes place) can be changed

Proposition: Let $\vec{a}, \vec{b}, \vec{c} \in V$ be arbitrary. Then

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = (\vec{b} \times \vec{c}) \cdot \vec{a}$$

$$= (\vec{c} \times \vec{a}) \cdot \vec{b}$$

and

(1.3) $$(\vec{a} \times \vec{b}) \cdot \vec{c} = \det\left(\begin{bmatrix} | & | & | \\ \vec{a} & \vec{b} & \vec{c} \\ | & | & | \end{bmatrix} \right).$$

Proof: Exercise ///

Corollary Combining (1.2) and (1.3), we have

(1.4) $$\det\left(\begin{bmatrix} | & | & | \\ \vec{a} & \vec{b} & \vec{c} \\ | & | & | \end{bmatrix} \right) = \epsilon_{i,j,k}\, a_i b_j c_k \qquad (3 \text{ sums})$$

Proposition (Triple Vector Product) Let $\mathcal{F} = \{\vec{e}_i\}$ be a frame

$$\vec{a} = a_i \vec{e}_i, \quad \vec{b} = b_j \vec{e}_j, \quad \vec{c} = c_k \vec{e}_k, \quad \vec{d} = d_\ell \vec{e}_\ell.$$

Then

(1.5) $$\vec{a} \times (\vec{b} \times \vec{c}) = \epsilon_{p,q,k}\, \epsilon_{i,j,k}\, a_q b_i c_j \vec{e}_p$$

If we set

$$\vec{d} = \vec{a} \times (\vec{b} \times \vec{c}),$$

then

$$d_p = \epsilon_{p,q,k}\, \epsilon_{i,j,k}\, a_q b_i c_j .$$

<u>Result</u> (1.1) (Epsilon-Delta Identities)

$$\epsilon_{p,q,s}\, \epsilon_{n,r,s} = \delta_{p,n}\delta_{q,r} - \delta_{p,r}\delta_{q,n}$$

and

$$\epsilon_{p,q,s}\, \epsilon_{r,q,s} = 2\delta_{p,r}$$

<u>Proof</u>: Exercise. ///

As a consequence of the last result, observe that

$$\begin{aligned}
\vec{a} \times (\vec{b} \times \vec{c}) &\overset{(1.5)}{=} \epsilon_{p,q,k}\, \epsilon_{i,j,k}\, a_q b_i c_j \vec{e}_p \\
&= (\delta_{p,i}\delta_{q,j} - \delta_{p,j}\delta_{q,i})\, a_q b_i c_j \vec{e}_p \\
&= a_q (\delta_{q,j} c_j)(\delta_{p,i} b_i)\vec{e}_p \\
&\qquad - a_q (\delta_{q,i} b_i)(\delta_{p,j} c_j)\vec{e}_p \\
&= (a_q c_q)\, b_p \vec{e}_p - (a_q b_q)\, c_p \vec{e}_p \\
&= (\vec{a}\cdot\vec{c})\,\vec{b} - (\vec{a}\cdot\vec{b})\,\vec{c}
\end{aligned}$$

That is

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a}\cdot\vec{c})\,\vec{b} - (\vec{a}\cdot\vec{b})\,\vec{c} \quad ///$$